Peradeniya

FACULTY OF SCIENCE - UNIVERSITY OF PERADENIYA

END SEMESTER EXAMINATION – SEMESTER II, 2010

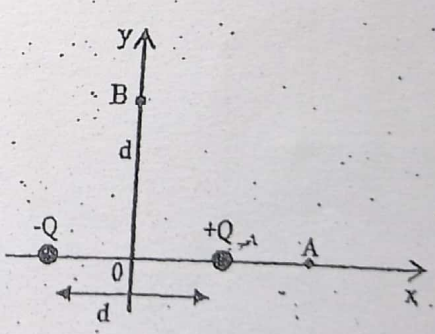
General Physics II- (PH102)-2 Credits

Time Allowed: Two hours.
Answer **Five** questions selecting at least **one question from each section.**
(Calculators will be provided)

Electron charge (e)	$= 1.60 \times 10^{-19}$ C
Electron mass (m_e)	$= 9.11 \times 10^{-31}$ kg
Planck's constant (h)	$= 6.63 \times 10^{-34}$ J s
Speed of light (c)	$= 3.00 \times 10^{8}$ m s^{-1}
Avogadro number (N_A)	$= 6.023 \times 10^{23}$ mol^{-1}
1 Electron Volt (1 eV)	$= 1.60 \times 10^{-19}$ J
1 Atomic mass unit (1 u)	$= 931.5$ MeV

Section-A

1. An electric dipole is placed on the x-axis with $+Q$ at $x = d/2$ and $-Q$ at $x = -d/2$ as shown in the figure. (Express your answers in terms of K, Q, d and x, where $K = 1/(4\pi\varepsilon_0)$).



(a) What is the electric potential $V(x)$ on the x-axis at a distance x from the origin where $x > d/2$?

(b) What is the electric potential on the x-axis at a point A at a distance $x = d$ from the origin ?

(c) Find the electric potential on the y-axis at a point B a distance $y = d$ from the origin.

(d) What is the x-component of the electric field $E_x(x)$ at a point on the x-axis at a distance x from the origin in the region $x > d/2$?
Deduce $E_x(x)$ when $x >> d/2$.

2. (a) State the Ampere's circuital law.
(b) Two very thin concentric hollow conducting cylinders with radii a and $3a$ carry uniformly distributed currents in opposite directions. The magnitude of the current

through the inner cylinder is I and the magnitude of the current through the outer cylinder is $3I$.

Find the magnetic field at distance r from the axis of symmetry in the three regions

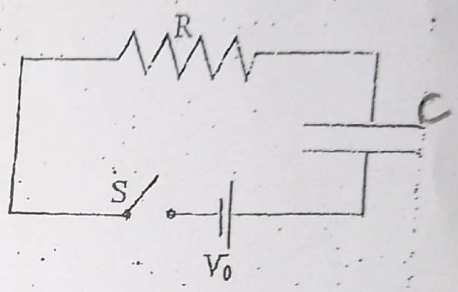
(i) inside the inner cylinder

(ii) between the two cylinders, and

(iii) outside the outer cylinder.

Indicate in a sketch, the directions of the currents in the two cylinders and the direction of the associated magnetic field found by you for the three regions.

3. A resistance R, a capacitor C, a switch S and a battery (with EMF V_0) are in series as shown in the figure. The switch is closed at $t = 0$. At that time, the capacitor holds no charge.



For $t > 0$:

(a) Write down the differential equation for the charge Q on the lower plate of the capacitor.

(b) Show, by substitution into your equation above, that $Q = CV_0\,(1 - e^{-t/\tau})$ is the correct solution, provided that τ is chosen properly. Determine τ.

(c) What is the current in the circuit at time t_1 ($t_1 > 0$)?

(d) Show that the energy stored in the capacitor at time t_1 is given by

$$U = \frac{1}{2}CV_0^2\left(1 - e^{-t_1/\tau}\right)^2$$

Section-B

4. (a) Einstein's equation on Photoelectric Effect can be written as:

$$h\nu = h\nu_o + \left(\frac{1}{2}mV^2\right)_{max}$$

where h is the Planck's constant and m and V are the mass and the velocity of the photo electrons emitted. Give the meaning of each term in this equation and show how it is used to explain the photoelectric effect.

(b) A metal whose work function is 3.0 eV, is illuminated by light of wavelength 3.0×10^{-7} m. Calculate (i) the threshold frequency, ν_o, (ii) the maximum energy of photoelectrons emitted, and (iii) the stopping potential.

CamScanner

5. (a) The energy of the electron in the nth Bohr orbit of a hydrogen-like atom is given by,

$$E_n = -Z^2\left(\frac{me^4}{8\varepsilon_o^2 ch^3}\right)\frac{hc}{n^2} = -13.6\,\frac{Z^2}{n^2}\ (\text{in eV})$$

where Z is the number of protons in the nucleus, m and e are the mass and the charge of the electron, h is the Planck's constant, c is the velocity of light, ε_o is the permittivity of free space and n is the Principal Quantum Number.

(b) Modify the above equation (without deriving it) to obtain an expression for the energy of the electron in the nth Bohr orbit of a singly-ionized helium atom (an atom from which one of the two electrons has been removed).

(c) Use your expression to calculate the energy (in eV) of the ground state ($n = 1$), and the first three excited states ($n = 2, 3, 4$) of the singly-ionized helium atom.

(d) Indicate these energies in an energy level diagram.

6. (a) The normal activity of a living tree containing radioactive carbon ($^{14}C_6$) is found to be 15 decays per minute per gram of carbon. An archeological wooden object from an ancient city gives 6 decays per minute per gram of carbon. If the half-life of $^{14}C_6$ is 5730 years, estimate the age of the wooden object.

(b) Consider the following "photo fission" reaction of $^{235}U_{92}$, induced by a 6 MeV γ–ray photon.

$$^{235}U_{92} + \gamma \rightarrow {}^{90}Kr_{36} + {}^{142}U_{56} + 3\,{}^{1}n_0$$

Find the energy released in the reaction.
You are given the following atomic masses:

$^{235}U_{92}$: 235.043915 u $^{90}Kr_{36}$: 89.91972 u

$^{142}U_{56}$: 141.91635 u $^{1}n_0$: 1.008665 u

Section -C

7. (a) Distinguish between a covalent bond, and a Van der Waals bond.

(b) Plot the "potential energy vs. distance of separation" curve for a diatomic molecule, identifying the important points of the curve.

(c) Use the above curve to explain thermal expansion, elastic modulus and melting temperatures of materials.

P.T.O

CamScanner

(d) Use the band theory of solids to explain the classification of solids into conductors, insulators, and semiconductors.

(e) Explain why semiconductors become better electrical conductors as the temperature increases.

(f) Use the idea of electronic configuration to explain how a "doped" semiconductor can become highly conducting.

8. (a) Show that the rotational inertia about the centre of mass of a diatomic molecule is given by $I = \mu R^2$, where the symbols have their usual meanings.

(b) Obtain an equation for the energy associated with the rotation of a diatomic molecule.

(c) Draw an energy level diagram showing two molecular vibrational energy levels and three rotational energy levels with allowed transitions.

(d) A hypothetical diatomic molecule is composed of two identical atoms each of mass 12 u. If the wavelength of the radiation emitted when rotational transition from $l = 2$ to $l' = 1$ occurs, is 2.13 mm, what is the rotational inertia? What is the distance of separation between the atoms?

****